IGCSE 0607 Practice Questions — E2.3

Algebraic fractions

Total Questions: 27 — **Total Marks:** 122 Marks

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  → **Paper 2 (Non-Calculator)** *(16 Qs)* · **Paper 4 (Calculator)** *(11 Qs)*

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1 Write as a single fraction in its simplest form.

$$\frac{2x}{5} - \frac{x-1}{3}$$

..... **[2]**

2 Write as a single fraction in its simplest form.

$$\frac{2x-1}{3}+\frac{x+4}{5}$$

..... **[2]**

3 Write as a single fraction in its simplest form.

$$\frac{3x}{4} - \frac{x-2}{5}$$

..... **[2]**

4 Write as a single fraction in its simplest form.

$$\frac{5}{x+2} + \frac{2}{3x-1}$$

..... **[3]**

5 Write as a single fraction in its simplest form.

$$\frac{x+3}{4} - \frac{2x-1}{5}$$

..... **[3]**

6 Write as a single fraction in its simplest form.

$$\frac{3x}{4} - \frac{x-2}{3}$$

..... **[3]**

7 Write as a single fraction in its simplest form.

$$\frac{2x}{3} - \frac{x-4}{5}$$

..... **[3]**

8 **(a)** Write as a single fraction in its simplest form.

$$\frac{2x}{3} + \frac{x-1}{4}$$

..... **[2]**

(b) Simplify fully.

$$\frac{15x^3y^2}{5xy^4}$$

..... **[2]**

9 **(a)** Simplify fully.

$$\frac{3x}{4} + \frac{2x}{5}$$

..... **[2]**

(b) Simplify fully.

$$\frac{5a^2b}{12} \div \frac{10ab^2}{9}$$

..... **[2]**

10 Simplify fully.

$$\frac{x^2 - 16}{x^2 - 5x + 4}$$

..... **[3]**

11 Simplify fully.

$$\frac{3x^2 - 12}{x^2 + 3x - 10}$$

..... **[4]**

12 **(a)** Write as a single fraction in its simplest form.

$$\frac{3}{x-2} - \frac{2}{x+5}$$

..... **[3]**

(b) Simplify fully.

$$\frac{x^2 - 16}{x^2 - 2x - 8}$$

..... **[3]**

13 **(a)** Simplify fully.

$$\frac{x^2 - 9}{2x^2 + 5x - 3}$$

..... **[3]**

(b) Write as a single fraction in its simplest form.

$$\frac{4}{x - 2} - \frac{3}{x + 1}$$

..... **[3]**

14 Simplify fully.

$$\frac{2x^2 + 5x - 12}{4x^2 - 9}$$

..... **[4]**

15 Simplify fully.

$$\frac{3x^2 - 5x - 2}{x^2 - 4} \div \frac{3x + 1}{x^2 - x - 2}$$

..... **[6]**

16 **(a)** Simplify fully.

$$\frac{2x^2 - 5x - 3}{4x^2 - 9}$$

..... **[3]**

(b) Write as a single fraction in its simplest form.

$$\frac{3}{x-2} + \frac{4}{x+1} - \frac{2x}{x^2 - x - 2}$$

..... **[4]**

17 Simplify fully.

$$\frac{x^2 - 9}{2x^2 + 5x - 3}$$

..... **[3]**

18 Write as a single fraction in its simplest form.

$$\frac{5}{x-2} - \frac{3}{2x+1}$$

..... **[3]**

19 Simplify fully.

$$\frac{2x^2 - 9x + 9}{2x^2 - 18}$$

..... **[4]**

20 Simplify fully.

$$\frac{2x^2 - 8}{x^2 + x - 6}$$

..... **[5]**

21 **(a)** Write as a single fraction in its simplest form.

$$\frac{x+3}{2x-1} - \frac{x-2}{x+4}$$

..... **[3]**

(b) Simplify fully.

$$\frac{x^2-16}{x^2+4x}$$

..... **[2]**

22 Simplify fully.

$$\frac{x^2 - 25}{x^2 - 3x} \div \frac{x^2 + 6x + 5}{2x - 6}$$

..... **[4]**

23 **(a)** Simplify fully.

$$\frac{2x^2 - 8x}{x^2 - 16} \div \frac{6x}{3x + 12}$$

..... **[4]**

(b) Write as a single fraction in its simplest form.

$$\frac{5}{x - 3} + \frac{2x + 1}{x^2 - 9}$$

..... **[3]**

24 **(a)** Write as a single fraction in its simplest form.

$$\frac{4}{x-3} - \frac{3}{2x-1}$$

..... **[3]**

(b) Simplify fully.

$$\frac{x^2-25}{2x^2-9x-5} \div \frac{x+5}{4x+2}$$

..... **[4]**

25 **(a)** Write as a single fraction in its simplest form.

$$\frac{3}{x-2} - \frac{x+5}{x^2-4}$$

..... **[3]**

(b) Simplify fully.

$$\frac{4x^2-25}{2x^2-5x} \times \frac{6x^3}{4x^2+10x}$$

..... **[4]**

26 **(a)** Write as a single fraction in its simplest form.

$$\frac{5}{x-2}+\frac{4}{2x+3}$$

..... **[3]**

(b) Simplify fully.

$$\frac{2x^2-5x-12}{4x^2-9}\div\frac{x-4}{2x-3}$$

..... **[4]**

27 **(a)** Simplify fully.

$$\frac{2x^2 + 5x - 3}{4x^2 - 1}$$

..... **[4]**

(b) Express as a single fraction in its simplest form.

$$\frac{5}{x - 3} + \frac{x + 2}{x^2 - 4x + 3}$$

..... **[4]**